INCH-POUND

MIL-D-0051256C(EA)
14 February 1992
USED IN LIEU
MIL-D-51256B
8 November 1984
SUPERSEDING
(See 6.9)

MILITARY SPECIFICATION

DETECTOR KIT, CHEMICAL AGENT, ABC-M18A2

This specification is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, based upon currently available technical information but it has not been approved for promulgated as a coordinated revision of MIL-D-51256B. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers one type of chemical agent detector kit

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6665

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SPECIFICATIONS

MILITARY

MIL-D-51082 - Detector Tubes, Chemical Agent
MIL-S-51189 - Substrate Solution and Dispenser

STANDARDS

FEDERAL

FED. TEST METHOD STD. NO. 101 - Test Procedures for Packaging Materials

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-77-2092 - Detector Kit, Chemical Agent, ABC-M18A2

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-77-2092 - Detector Kit, Chemical Agent, ABC-M18A2

(Copies are available from Commander, U.S. Army Information Systems Command, ATTN: ASQNC-ARI-CT, Rock Island, IL 61299-7210.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-77-2092 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

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3.1.2 Components. All components of the detector kit shall conform to the specifications and drawings listed on Drawing 5-77-2092 and subsidiary drawings.

3.1.2.1 Sodium hydroxide bottle (Drawing 5-77-2100-1). The sodium hydroxide bottle shall contain 0.68 ± 0.04 gram (g) of ACS grade sodium hydroxide when tested as specified in 4.4.6.1.

3.1.2.2 Tris buffer bottle (Drawing 5-77-2100-2). The tris buffer bottle shall contain at least 8 milliliters (mL) of 0.05 molar tris (hydroxymethyl) aminomethane solution with a pH of 8.0 ± 0.1 when tested as specified in 4.4.6.2.

3.2 Assembly. The kit shall be assembled in accordance with Drawing 5-77-2092.

3.2.1 Discard date of assembled kit. The discard date of an assembled kit shall be 3 years after the earliest manufacturing date of the following components:

a. Detector tubes (MIL-D-51082)
b. Belt, detector tickets (5-77-808)
c. Substrate solution and dispenser (MIL-S-51189)

The assembled kit shall be submitted to the Government at least 2 years and 8 months before the discard date of the kit.

3.3 Performance.

3.3.1 Drop stability.

3.3.1.1 Case. The detector kit case shall not crack or break when tested as specified in 4.3.3.1.

3.3.1.2 Contents. The detector kit components shall not crack or break when tested as specified in 4.3.3.2.

3.4 First article. A sample shall be subjected to the first article inspection (see 6.3) in accordance with 4.3.

3.5 Workmanship. The detector kit, components, and sub-assemblies shall be free from defects and from contamination by foreign matter such as dirt, grease, and oil.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the

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performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.4.) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Sample. The first article sample shall consist of eight detector kits manufactured and packaged (when applicable) using the same methods, materials, equipment, and processes as will be used during regular production and three empty unit pack containers. Empty

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unit pack containers shall not be part of the first article sample when packaging is for inter-plant shipment. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.3.2 Inspections to be performed. As determined by the Government, the sample first article items may be subjected to the examinations and tests specified in 4.3.3, 4.4.5, 4.4.6 and 4.4.7 and be inspected for compliance with any or all of the requirements of the applicable drawings and special packaging instructions. However, unless otherwise specified the first article sample shall be examined for the characteristics specified in 4.4.5; three packaged detector kits from the first article shall be tested in accordance with 4.4.7.1 and the three empty unit packed containers shall be tested in accordance with 4.4.7.2. The first test conducted on the detector kits shall be 4.3.3.1, using three kits from the first article sample. The remaining five kits shall be tested in accordance with 4.3.3.2, 4.4.6.1 and 4.4.6.2, in that order.

4.3.3 Drop stability tests. Drop stability tests shall be conducted as follows:

4.3.3.1 Case. Bring each sample detector kit to a temperature of –40° ± 5°F and drop the kit twice free-fall, from a height of 6.0 (+0.5 or –0) feet onto a 1/2-inch thick plywood sheet on a concrete floor. After the drops, examine the case for evidence of cracking or breaking.

4.3.3.2 Components. Sample kits assembled in accordance with Drawing 5-77-2092 shall be dropped onto a 1/2-inch thick plywood sheet resting on concrete. Each kit shall be dropped three times in a random orientation from a height of 5 (+0.5 or –0) feet. The components of the kit shall be examined for any signs of breakage. Special attention shall be given to breakage of the detector tubes. A maximum of 5 broken detector tubes shall be allowed in each kit as a result of the drop test.

4.3.3 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the items produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions, and at essentially the same time. However, no more than one lot of chemical components shall be represented in any one lot of kits.

4.4.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4.4.5 and table I. Samples shall be selected at random.

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TABLE I. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

4.4.3 Inspection procedure. Every item in the lot shall be inspected for critical characteristics. The lot represented shall be rejected when nonconformance to a critical characteristic is found. Sample items and, when applicable, the packaging thereof shall be examined and tested in accordance with the classification of characteristics in 4.4.5. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.4.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product or whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its

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intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.4.5 Classification of characteristics. Quality conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. Unless otherwise specified, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table I for selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 5-77-2092
4.4.5(a)	Detector Kit, Chemical Agent, ABC-M18A2		NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Weight of sodium hydroxide in bottle	5 bottles, acc 0, rej 1	3.1.2 1	4.4.6.1*
102	pH of tris buffer solution	5 bottles; acc 0, rej 1	3.1.2.2	4.4.6.2*
103	Component present and correctly assembled	Table I, Level VIII	3.2	VI
104	Interior and exterior marking present, correct, and legible	Table I, Level VIII	3.2	VI
105	Rubber tubing correctly assembled in ticket adapter (Dwg 5-77-792)	Table I, Level VIII	3.2	VI
106	Discard date of kit and components as specified	Table I, Level VIII	3.2.1	VI
Minor				
201	Components properly placed inside kit	Table I, Level IX	3.2	VI
202	Hardware coating as specified (Dwg 5-77-168)	Table I, Level IX	3.1.1	VI
203	Workmanship	Table I, Level IX	3.5	VI

NOTES:

VI - Visual inspection
* Destructive test

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.5(b)	Packaging		SPI P5-77-2092
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Unit pack container leakage	Table I, level X	5.1	4.4.7.1
102	Unit pack container seam	5; Acc 0, Rej 1*	5.1	4.4.7.2**
103	Components present and correctly assembled	Table I, level X	5.1	VI
104	Item completely clean and dry prior to unit packing	Table I, level X	5.1	VI
105	Contents of container as specified	Table I, level X	5.1	VI
106	Unit pack container and closure as specified	Table I, level X	5.1	VI
107	Fiberboard pads present and as specified	Table I, level X	5.1	VI
108	Packing container and closure as specified	Table I, level X	5 1	VI
109	Supplemental container and closure as specified	Table I, level X	5 1	VI
110	Marking present, correct and legible	Table I, level X	5.1	VI

NOTES:
This paragraph is not applicable to packaging for interplant shipment.
* Samples shall consist of closed empty unit pack containers
** Destructive test
VI - Visual inspection

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4.4.6 Tests. Tests shall be conducted as follows:

4.4.6.1 Weight of sodium hydroxide in bottle (Dwg 5-77-2100-1). Determine the weight of sodium hydroxide in each bottle in the sample by titrating with a standard acid solution.

4.4.6.2 pH of tris buffer solution (Dwg 5-77-2100-2). Determine the volume and the pH of the tris buffer solution in each bottle in the sample using a graduated glass cylinder and a suitable pH meter, accurate to 0.1 pH units.

4.4.7 Packaging tests.

4.4.7.1 Unit pack bag container leakage. The leakage of the unit pack bag containing the M18A2 kit shall be determined by subjecting the sample to the hot water technique Method 5009.3 of FED. TEST METHOD STD. NO. 101.

4.4.7.2 Unit pack container seam. The closure seam of the empty unit pack container shall be prepared under the same conditions used for unit pack assemblies. The closure seam of the empty unit pack container shall be tested in accordance with method 2024 of FED. TEST METHOD STD. NO. 101, except that the static load shall be 45 (+5 or -0) ounces and a 5 percent reduction in the static load shall be permitted when the room temperature exceeds 90°F in the test area.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with SPI P5-77-2092.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This kit is intended for use in detecting chemical agents in field concentrations.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 1 and 2.1.2)
(c) Level of packing required (see 5.1)
(d) Shelf life code
(e) First article:

(1) Time allowed for contractor submission of samples for Government test and

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evaluation after award of contract when testing is performed by the Government.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the contractor whether or not to proceed with production

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first 500 production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternative bids unless specifically requested to do so in the solicitation.

6.4 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.5 Material safety data sheets. The Contracting Officers will include in all solicitations the following clauses relating to the preparation and submission of Material Safety Data Sheets (MSDS).

(a) FAR 52.223-3, Hazardous Material Identification and Material Safety Data

(b) DFARS 252.223-7001, Hazard Warning Labels

(c) Submission of MSDS and warning labels will be made to the Government mailing addresses as listed in FED-STD-313 and any other local organization as required by the PCO.

6.6 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Related documents. The following documents support the ABC-M18A2 detector kit:

Packet, Sodium Pyrophosphate Peroxide	MIL-P-50014
Tablets, Compound 34	MIL-T-51023
Detector Tubes, Chemical Agent	MIL-D-51082
Detector Ticket, Anticholinesterase Agent	MIL-D-51083

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Substrate Solution and Dispenser	MIL-S-51189
Paper, Chemical Agent Detector, VGH, ABC-M8; Booklet of	MIL-P-51409

6.8 Subject term (key word) listing.

Sodium hydroxide
Toxic agent detector

6.9 Supersession data. This revision supersedes Interim Amendment 2 dated 27 February 1989.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No.6665-A622

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